

05 - Starting, Continuing, and Stopping Progesterone

Companion reading handout for simultaneous listening.

Course: Medical Prevention of Spontaneous Early Delivery

Here is the apparent contradiction.

Why do many protocols continue vaginal progesterone until thirty-four, thirty-six plus six, or thirty-seven weeks, while the strongest recommendations resist starting progesterone after twenty-four weeks?

If progesterone is still being given at thirty-four weeks, why not begin it at thirty-four weeks?

This is exactly the kind of question that should make a resident pause. It sounds like a contradiction only because two different questions have been collapsed into one.

Starting treatment and continuing treatment are not the same evidence claim.

Starting treatment asks: if we identify the patient now, at this gestational age, can a new progesterone regimen still change the trajectory toward early delivery?

Continuing treatment asks: if treatment was started in the evidence window, should the evidence-based regimen be completed until the protocol stop point?

Those are different questions.

Trial Stop Points

The randomized short-cervix trials generally began in the mid-trimester. Fonseca screened at twenty to twenty-five weeks, randomized women with a cervix fifteen millimeters or less, and gave two hundred milligrams vaginal progesterone nightly from twenty-four weeks to thirty-three plus six. PREGNANT enrolled women at nineteen to twenty-three plus six weeks, used ninety milligrams vaginal gel daily, and continued until thirty-six plus six, rupture of membranes, or delivery. OPPTIMUM started around twenty-two to twenty-four weeks and continued to thirty-four weeks.

So when a patient started treatment at twenty-one or twenty-two weeks and is still taking it at thirty-two weeks, the clinician is not claiming that new initiation at thirty-two weeks has been proven effective. The clinician is completing a regimen that began when the patient resembled the evidence population.

Now look at late initiation.

Timing Gradient

The Einum study from two thousand twenty-five is valuable because Norwegian guidelines allowed treatment initiation across a wide interval, from sixteen to thirty-one weeks, in pregnancies with a short cervix. The study divided diagnosis and treatment timing into sixteen to twenty-one weeks, twenty-two to twenty-seven weeks, and twenty-eight to thirty-one weeks.

In the full sample, progesterone was associated with lower risk of birth before twenty-eight weeks: zero point eight percent versus three point four percent, adjusted relative risk zero point two five. But the timing gradient is the teaching point.

Among mothers diagnosed from sixteen to twenty-one weeks, the adjusted relative risk for birth before twenty-eight weeks was zero point one three, and for birth before thirty-four weeks it was zero point two seven. That is the strongest signal.

From twenty-two to twenty-seven weeks, the estimates still pointed toward possible reduction, but they were imprecise: adjusted relative risk zero point four two for under twenty-eight weeks and zero point six eight for under thirty-four weeks, with wide confidence intervals.

From twenty-eight to thirty-one weeks, there was no apparent benefit. For birth before thirty-four weeks, the adjusted relative risk was one point three zero. For birth before thirty-seven weeks, it was one point one four.

The memory is not just "late is bad." The memory is effect-gradient by timing. Sixteen to twenty-one weeks: strongest signal. Twenty-two to twenty-seven: uncertain and imprecise. Twenty-eight to thirty-one: no apparent benefit.

The biology makes this plausible. If the goal is to prevent very early delivery, then a patient first detected at twenty-nine or thirty weeks has already passed much of the extreme-preterm interval. The process may also be closer to established cervical remodeling, mechanical stress from the growing fetus, or functional progesterone withdrawal that exogenous progesterone cannot easily reverse.

Late Diagnosis Changes the Endpoint

Late diagnosis also changes the endpoint. A patient diagnosed at thirty weeks cannot have a birth before twenty-eight weeks. That endpoint is no longer available. The clinical question becomes different: can we prolong pregnancy, and by how much?

That is why the late-start studies that measure diagnosis-to-delivery interval are so important.

Late-Start Evidence

Dominsky and colleagues studied singleton pregnancies with no prior preterm birth and cervical length less than twenty-five millimeters diagnosed between twenty-four plus zero and thirty-three plus six weeks. In a one-to-one matched analysis, vaginal progesterone did not reduce spontaneous preterm birth at any threshold. Under thirty-seven weeks was twenty-nine point three percent with progesterone versus twenty-six point two percent without. Under thirty-four weeks was seven point zero percent versus eight point two percent. Under thirty-two weeks was three point nine percent versus four point seven percent. Under twenty-eight weeks was zero point four percent versus one point six percent. The interval from diagnosis to delivery was also essentially the same: forty-eight point six versus forty-nine point five days.

That last number is memorable because it answers the continuous-outcome question. If you start late in this low-risk no-prior-preterm-birth phenotype, the study did not show that you gain time.

The two thousand twenty-six American Journal of Obstetrics and Gynecology Maternal-Fetal Medicine abstract asks a similar question in asymptomatic singleton pregnancies first found to have a cervix twenty-five millimeters or less after twenty-four weeks. It also found no reduction. Birth before thirty-seven weeks was twenty-four point six percent with progesterone versus nineteen point one percent with expectant management. Birth before thirty-four weeks was three point one percent versus five point six percent. Time-to-delivery analysis showed no difference, with a hazard ratio of one point two eight and a confidence interval from zero point six five to two point five four.

Use that abstract cautiously because it is an abstract source, not a full-text conversion in the corpus. But its result supports the same teaching point: late initiation in an asymptomatic late short-cervix phenotype has not shown a clear time-to-delivery benefit.

Now add the high-risk prior-preterm-birth late-short-cervix population.

High-Risk Late Short Cervix

Luxembourg and colleagues studied women with a history of preterm birth and cervical length twenty-five millimeters or less diagnosed after twenty-four weeks. This is a different phenotype. The study suggested lower spontaneous preterm birth before thirty-seven weeks, thirty-one point two percent versus forty-two point eight percent, and no births before thirty-four weeks in the treated group versus seven point one percent in the untreated group. But the adjusted hazard ratio for delivery before thirty-seven weeks was not statistically significant, one point five three nine with a confidence interval from zero point seven nine to three point zero two. The study is retrospective and vulnerable to confounding.

So do not generalize it as proof. Store it as hypothesis-generating in a high-risk phenotype.

Resolving the Paradox

Now return to the original paradox.

Continuing progesterone until thirty-four or thirty-six plus six weeks is not the same as starting progesterone at thirty-four weeks. Continuation belongs to a regimen that began in the evidence zone. Late initiation asks whether a new treatment can still alter biology and gain time after the risk signal appears late.

That is the memory structure.

Twenty-four weeks is not a magic doorway in the cervix. It is an evidence boundary. Thirty-four or thirty-six plus six weeks is often a protocol stop point. Do not confuse the evidence boundary with the stop point.

A careful physician can say: "We have good evidence for starting vaginal progesterone in the mid-trimester short-cervix phenotype. If it was started then, continuing to the regimen's stop point is protocol completion. But if the cervix is first found short after twenty-four weeks, especially after twenty-eight weeks, the evidence for starting is weaker, observational, and in several studies negative for prolonging pregnancy."

That sentence is harder than a rule. It is also the truth.